What is Solace Architect?

Solace Architect is an AI-powered solution architect that takes you from business requirements to a complete Solace architecture — and the business case to back it up.

You describe your systems and goals. Solace Architect runs a structured engagement: discovery, design, reviews, validation. What comes out the other end is a technical blueprint your engineering team can build from, and an executive summary your leadership team can fund with.

The problem it solves

Designing an event-driven architecture on Solace involves dozens of decisions: broker type, topic taxonomy, protocol selection, mesh topology, HA/DR strategy, security posture, migration path, governance model. Each decision depends on the others, and getting them wrong is expensive to fix later.

Today, these decisions live in the heads of senior architects. The knowledge is real but scarce — and an engagement that should take days stretches into weeks of meetings, slide decks, and back-and-forth.

Solace Architect compresses that timeline. It brings the structured thinking of an experienced Solace architect into every engagement, every time — consistent, thorough, and grounded in Solace documentation.

What it does

A Solace Architect engagement has three phases:

1. Understand

Everything starts with discovery. You describe the systems that need to communicate, the protocols they speak, the event types that flow between them, and the requirements that constrain the design: latency targets, delivery guarantees, topology, compliance, scale.

Solace Architect asks the right questions, identifies what matters, and captures it in a structured discovery brief. No detail is assumed — every architectural decision traces back to a stated requirement or an explicit design choice.

For teams that prefer to collect requirements offline, the intake template (available as a Word document, YAML, or Markdown) lets stakeholders fill in what they know before the engagement begins. What they leave blank, Solace Architect follows up on.

2. Design

With discovery complete, Solace Architect works through each design dimension:

- **Topic taxonomy** — the naming structure that organizes every event in the system
- **Broker selection** — cloud-managed, self-hosted software, or appliance, with sizing
- **Protocol selection** — which protocol fits each system's needs and constraints

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- **Mesh topology** — how brokers connect across sites, regions, and clouds
- **HA and DR** — what fails, how it recovers, and what the business loses in between
- **Integration** — which Micro-Integrations connect external systems to the event mesh
- **Migration** — if moving from another platform, the phased path to get there
- **Event Portal governance** — how events are cataloged, discovered, and promoted

Each step produces a concrete artifact: a topology document, a protocol map, a migration plan. Every recommendation is grounded in Solace platform documentation — not invented, not borrowed from other vendors, not hallucinated.

At each decision point, Solace Architect presents the options, the tradeoffs, and a recommendation. You choose. The design accumulates your decisions into a coherent architecture.

3. Validate and deliver

Before anything is finalized, four independent reviews check the design:

- **Architecture review** — structural soundness, antipattern detection
- **Operations review** — day-two readiness, monitoring, upgrade paths
- **Security review** — authentication, encryption, access control, compliance
- **Developer experience review** — SDK choices, schema governance, onboarding

Findings feed back into the design. What remains is validated, cross-checked, and assembled into two deliverables:

The technical blueprint — a deployment-ready specification covering every layer of the architecture: infrastructure, topic design, queue bindings, security configuration, monitoring setup, and CI/CD integration.

The executive summary — the business case for the architecture: what it enables, what it costs, what it saves, and the ROI framework with sensitivity analysis that lets leadership pressure-test the numbers.

How it works in practice

There are two ways to run an engagement:

The three-command path — for most projects, three commands cover the full arc:

1. `/solace-discovery` — describe your systems and goals
2. `/solace-plan` — Solace Architect picks the right skills, runs them in order, and pauses only for design decisions
3. `/solace-projects` — track progress, review artifacts, export the report

Individual skills — for teams that want to run specific design steps independently, each skill is available as a standalone command. Re-run topic design after requirements change. Add a security review to an existing project. Pick up where you left off.

Every engagement produces a project folder with all artifacts, decisions, and a timing log. The project dashboard gives you a single view of everything produced — and the HTML report packages it for stakeholders who were not in the room.

The business case

A good architecture that never gets funded is the same as no architecture. Solace Architect closes that gap by producing the business case alongside the technical design — not as an afterthought, but as a first-class deliverable.

Executive summary — a narrative written for leadership, not engineers. It frames the architecture in business terms: what capability it unlocks, what risk it retires, what operational cost it eliminates. The format is ready to drop into a funding request or a steering committee deck.

ROI framework — a structured model that quantifies the value of the architecture across five dimensions:

- *Development velocity* — time saved by standardized event patterns and pre-built Micro-Integrations versus custom point-to-point integration
- *Operational efficiency* — reduction in incident response time, lower monitoring overhead, fewer manual interventions
- *Infrastructure optimization* — right-sized broker selection, avoided over-provisioning, cloud cost management
- *Risk reduction* — quantified cost of downtime avoided through HA/DR design, security posture improvements, compliance coverage
- *Time to market* — faster delivery of new event-driven capabilities using the designed platform versus building from scratch

The ROI model includes a sensitivity analysis with adjustable parameters — adoption rate, implementation timeline, cost assumptions — so leadership can pressure-test the numbers against their own expectations. A combined scenario shows how multiple what-if adjustments compound, with delta indicators showing the shift from baseline projections.

The result: engineering gets a blueprint they can build from, and leadership gets a business case they can fund with. Same engagement, both audiences served.

What it is not

Solace Architect does not deploy infrastructure. It does not write application code. It does not replace the judgment of an experienced architect — it augments it.

It is a design tool, not a runtime tool. The output is a blueprint: precise enough to build from, reviewed enough to trust, and documented enough to hand to someone who was not part of the conversation.

Who it is for

- **Solution architects** running Solace engagements — faster time-to-design, consistent quality, nothing missed
- **Engineering teams** evaluating Solace — structured discovery that surfaces

the right questions before the first line of code

- **Technical leaders** building the business case — ROI framework and executive summary that speak the language of funding decisions

- **Existing Solace customers** extending their platform — migration planning, new use case design, architecture review of what is already running

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Built on what Solace knows

Every recommendation Solace Architect makes is grounded in Solace documentation: `docs.solace.com`, the Solace Agent Mesh project, the Integration Hub catalog, and published reference architectures. When a capability is not documented, Solace Architect says so — it does not guess.

The naming is precise. The terminology is exact. The antipatterns are cataloged and checked against. This is not a general-purpose AI giving generic advice about messaging — it is a Solace-specific tool that knows where the platform shines and where the sharp edges are.